

*MA.8.AR.3.2***Benchmark**

MA.8.AR.3.2 **Given a table, graph or written description of a linear relationship, determine the slope.**

Benchmark Clarifications:

Clarification 1: Problem types include cases where two points are given to determine the slope.

Clarification 2: Instruction includes making connections of slope to the constant of proportionality and to similar triangles represented on the coordinate plane.

Connecting Benchmarks/Horizontal Alignment

- MA.8.AR.3.3, MA.8.AR.3.5
- MA.8.F.1.1
- MA.8.GR.2.4
- MA.8.DP.1.3

Terms from the K-12 Glossary

- Slope

Vertical Alignment**Previous Benchmarks**

- MA.7.AR.3.2

Next Benchmarks

- MA.912.AR.2.2

Purpose and Instructional Strategies

In grade 7, students determined the constant of proportionality in a proportional relationship. In grade 8, students are determining the slope of a linear relationship from a given table, graph, or written relationship. In Algebra 1, students will write a two-variable linear equation from a graph, written description, or a table to represent relationships between quantities in mathematical and real-world context.

- Students identified the unit rate or the constant of proportionality in prior grade levels. This benchmark is the first one that references the slope, which represents a constant rate of change. This is not the same as a constant of proportionality unless the relationship goes through the origin.
- Instruction includes interpreting the meaning and value of slope in real-world context.
- Understanding slope can be introduced through a graph and the change in value of the y and the x .
- To introduce the concept to students, use at least two points on a graph in quadrant one.
- Instruction includes using a variety of vocabulary to make connections to real-world concepts and future courses. To describe the slope, one can say either “the vertical change divided by the horizontal change” or “rise over run.”
- Students should have experience utilizing a slope formula to determine the slope between two points on a line.
 - Slope of a line can be found by the expression $m = \frac{y_2 - y_1}{x_2 - x_1}$, where (x_1, y_1) and (x_2, y_2) are two different points on the line.

Common Misconceptions or Errors

- Students may invert the x - and y -values when calculating slope. To address this misconception, students should represent the relationship visually.

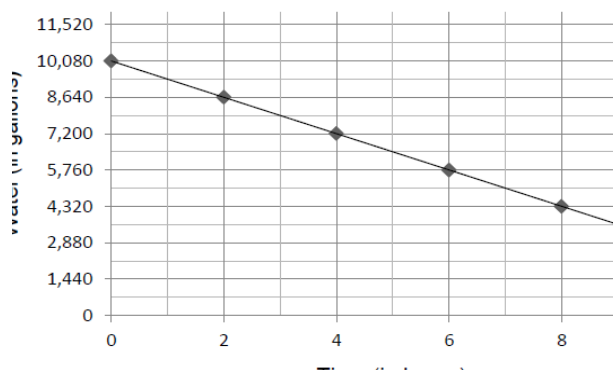
Strategies to Support Tiered Instruction

- Teacher supports students who invert the x - and y -values when calculating slope by using real-world problems that students can relate to and helping students represent the relationship visually.
- Instruction includes providing students with graph paper with grid lengths larger than 1 centimeter and using appropriate scaling of the axes to allow for students to see the unit rate more easily.

Instructional Tasks

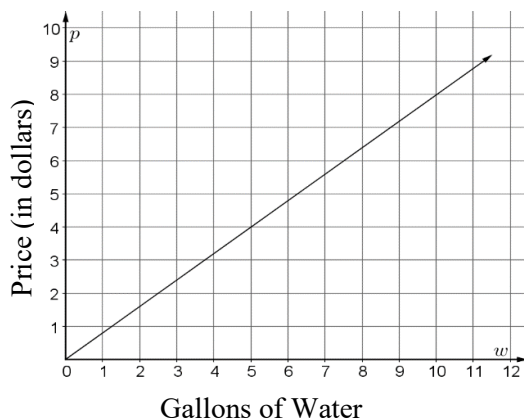
Instructional Task 1 (MTR.6.1)

Mr. Elliot needs to drain his above ground pool before the winter. The graph below represents the relationship between the number of gallons of water remaining in the pool and the number of hours that the pool has drained. Determine the slope and explain what it means in this situation.



Instructional Task 2 (MTR.4.1, MTR.7.1)

Jack and Jill are selling gallons of water that are sold in different size pails. Jack charges \$1.75 for every 2 gallons of water a pail holds. Additionally, he charges a \$2 service fee. Jill's prices can be modeled with the graph shown.



Part A. Identify the slope of Jack's relationship. Explain what it means.

Part B. Identify the slope of Jill's graph and explain what it means.

Part C. Graph Jack's prices on the same graph as Jill.

Part D. Whose has the better deal, Jack or Jill? Explain.

Instructional Items

Instructional Item 1

Cameron joins a local gym. There is an initial fee of \$25 and then a fee of \$5 per month. Determine the slope. After 12 months how much money has Cameron paid to the gym?

Instructional Item 2

A linear relationship is given in the table below. Determine the slope of the relationship.

x	-1	0	1	2	3
y	-18	-9	0	9	18

**The strategies, tasks and items included in the BIG-M are examples and should not be considered comprehensive.*